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A Deep Learning and Handcrafted Feature Fusion Framework for Automated Diabetic Retinopathy Grading from Retinal Fundus Images

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Abstract

Diabetic retinopathy (DR) is a leading cause of preventable blindness, especially in low-resource settings such as Bangladesh, creating urgent need for scalable medical image classification tools for early detection and progress towards SDG-3. This study proposes and evaluates a deep learning and handcrafted feature fusion framework for five-class DR grading (no, mild, moderate, severe, proliferative) from retinal fundus images using a blended dataset constructed from APTOS 2019 and IDRiD. After duplicate removal, we apply preprocessing with resizing, contrast enhancement, circular cropping, and color normalization. APTOS images are augmented to mitigate class imbalance and split into training, validation, and test sets in a 70:15:15 ratio. IDRiD images follow the original train-test split; the train set is split 85:15 into training and validation, and augmentation is applied only to training images. The blended sets are APTOS train plus augmented IDRiD train, APTOS validation plus IDRiD validation, and APTOS test plus preprocessed IDRiD test, yielding 6,740 images (5,514 training, 592 validation, 634 test). We extract deep features from pretrained EfficientNet-B3, ResNet101, DenseNet121, and MobileNetV2 and handcrafted descriptors (LBP, HOG, SIFT), and investigate three baseline strategies: frozen deep features, handcrafted features alone, and early feature fusion via concatenation of deep and handcrafted descriptors. We then fine-tune each CNN using head training, gradual unfreezing, and full-network optimization, and classify the resulting representations using SVMs, XGBoost, and ExtraTrees. Performance on the blended test set is measured using accuracy, precision, recall, macro F1-score, area under the ROC curve, and quadratic weighted kappa. The best-performing configuration, based on EfficientNet-B3 features with an SVM classifier, achieves a test accuracy of 72.7%, macro F1-score of 58.4%, area under the ROC curve of 0.90, and quadratic weighted kappa of 0.59, with variability over repeated runs summarized using standard deviations and 95% confidence intervals. These findings indicate that feature fusion, multi-source training, and rigorous statistical evaluation can underpin scalable DR screening tools for resource-limited healthcare systems and large-scale screening programs.